

Project Activity
Network Coding for Wireless Applications: A
Review
Wireless and Mobile Network



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INTRODUCTION

WHAT IS NETWORK CODING?

Fundamentally network coding constitutes the application of mathematical operations (often from finite fields) on packets at intermediate nodes. Instead of simply forwarding individual packets, a node can combine several incoming packets into an outgoing packet through operations such as XOR. The destination, while arriving at these coded packets, will be able to decode them to recover the original information if enough coded packets are received. This can be considered analogous to solving a system of linear equations.

What is Network Coding?

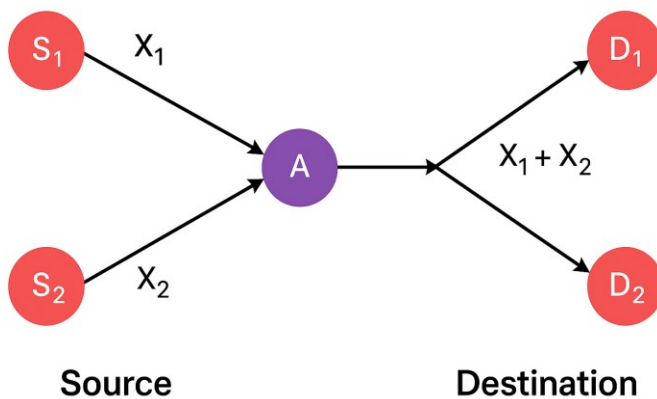


Figure 1:WHAT IS NETWORK CODING

There are two main types of network coding:

- **ALGEBRAIC NETWORK CODING:**

This is the most familiar scenario for which the linear combination of packets is done over a finite field. This provides a theoretical guarantee for attaining the max-flow min-cut capacity of a network.

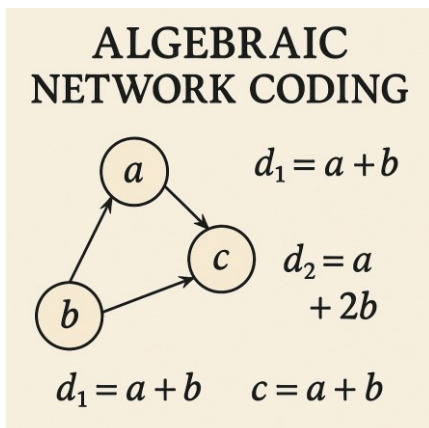


Figure 2:ALGEBRAIC NETWORK CODING

- **PRACTICAL NETWORK CODING(e.g., XOR-BASED NETWORK CODING):**

Using an XOR operation, this is a simpler and more widespread implementation. Algebraic network coding may not always achieve the theoretical capacity for which it was designed, but the low computational complexity gives it an advantage with high practicality in applications.

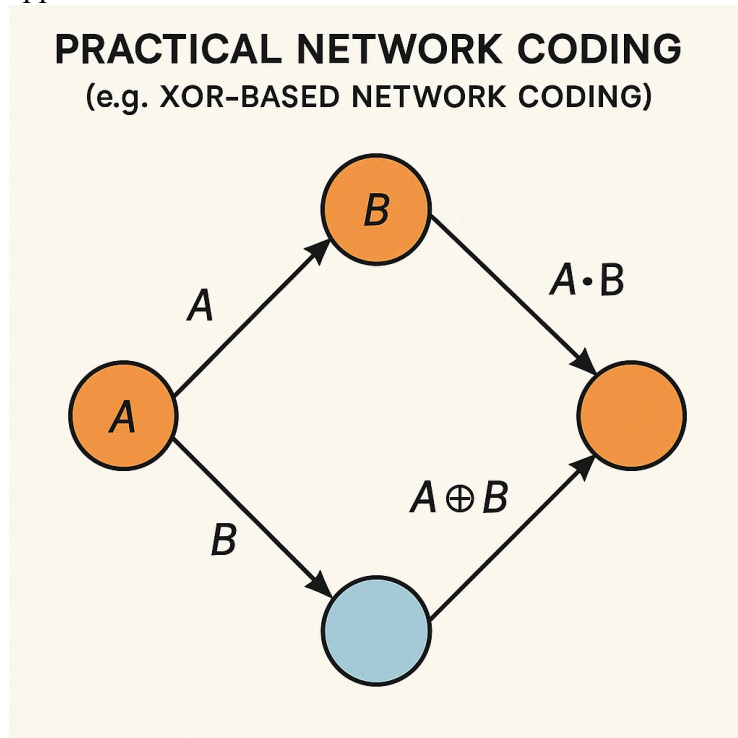


Figure 3: PRACTICAL NETWORK CODING

WHY NETWORK CODING?

Network coding addresses several inherent challenges in wireless communication:

- **INCREASED THROUGHPUT AND CAPACITY:**

By enabling intermediate nodes to combine information, network coding can effectively increase the overall throughput of a network, especially in multicast or several unicast flow scenarios. This theoretical maximum provision can achieve minimal-cut capacity, which is often not possible with traditional routing.

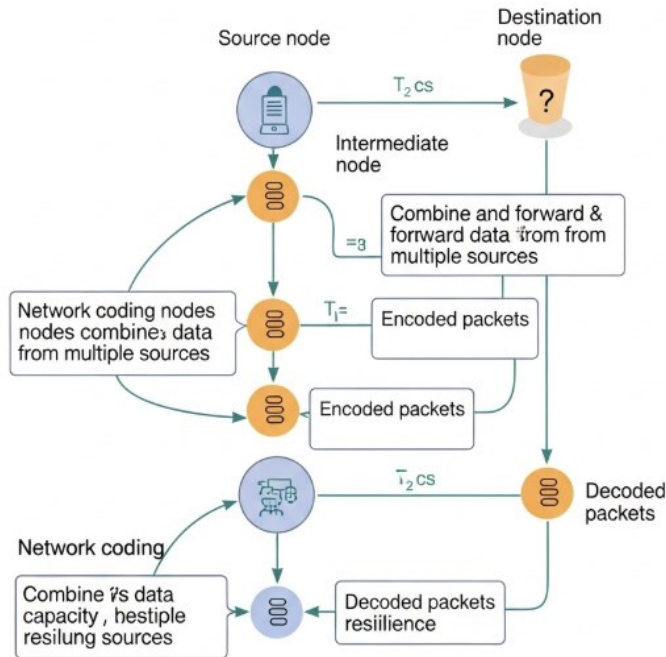


Figure 4: INCREASED THROUGHPUT AND CAPACITY

- IMPROVED ROBUSTNESS AND RELIABILITY:**

In front of packet damage, which is common in the wireless environment due to disappearance, intervention and mobility, the network coding provides the underlying variety. A lost coded packets can often be compensated by other obtained coded packets, which can reduce the requirement of retransmission and improve reliability.

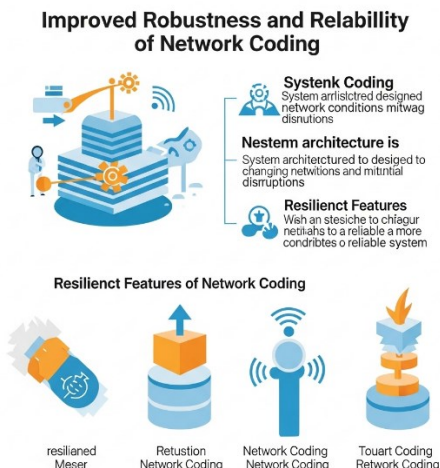


Figure 5: IMPROVED ROBUSTNESS AND RELIABILITY

- ENHANCED SECURITY:**

Network coding can be used to improve safety, which can improve safety by making it hard for eavesdroppers to decode information without sufficient number of coded packets. It can also be integrated with cryptographic techniques for stronger safer communication.

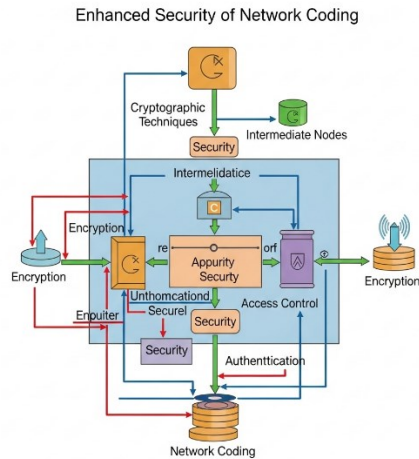


Figure 6:ENHANCED SECURITY

- **REDUCED ENERGY CONSUMPTION:**

By reducing the number of required broadcasting and retransmission, the network coding can contribute to significant energy savings to battery-operated wireless devices, expanding the network lifetime.

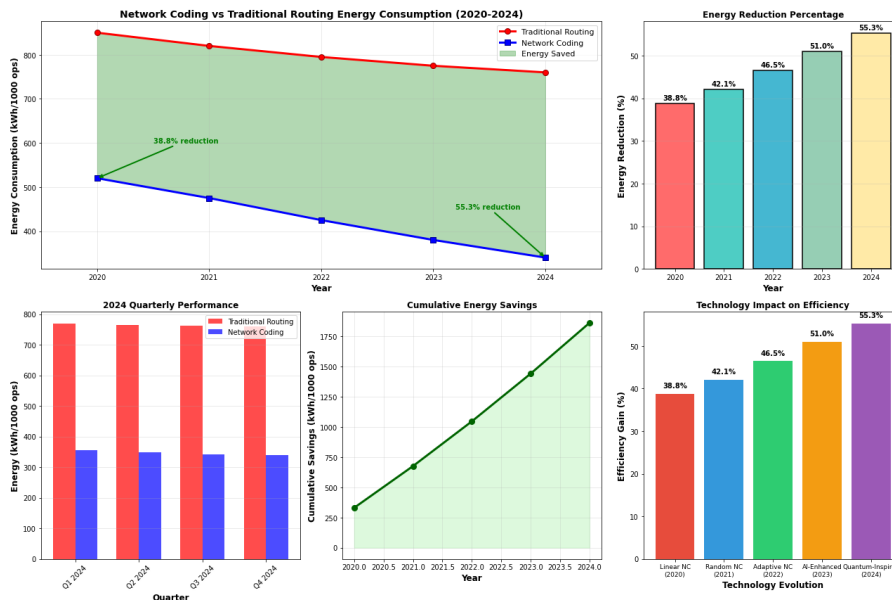


Figure 7:REDUCED ENERGY CONSUMPTION

- **EFFICIENT RESOURCE UTILIZATION:**

Network coding can adapt to the use of rare wireless bandwidth by combining data from intelligence, which can lead to more efficient spectrum.

- **DECENTRALIZED OPERATION:**

Many network coding schemes can function in a decentralized manner, making them well-adapted to dynamic and self-organizing wireless networks without a central controller.

HOW DOES NETWORK CODING WORKS?

In practice, **Linear Network Coding (LNC)** is the most commonly adopted form. Nodes perform operations like:

$$Y=aX_1+bX_2+...+nX_k$$

Where a, b , ..., n are coefficients from a finite region and X_i are upcoming packets. At the destination, gaussian eradication or matrix is used to decode the original packets. Random linear network coding (RLNC), Physical Layer Network Coding (PLNC), and coded cooperation neck to adapt to specific wireless references, adapt to vehicles ad hoc network (VANETS), IOT network and performance in satellite communications.

CONCLUSION:

In conclusion, the network coding is a revolutionary approach to data forwarding that introduces algebraic manipulation of packets at intermediate nodes, enabling intelligent and efficient data dissemination. Techniques like Algebraic Network Coding and XOR-based Practical Network Coding enable systems to achieve theoretical maximum network capacity while maintaining low computational complexity, especially in wireless and mobile environments. Network coding's core advantage lies in enhanced throughput, robustness, and efficiency. It allows packets to be mixed and decoded at the receiver using linear algebra techniques, reducing transmissions required for complete information delivery. This improves reliability under packet loss and mobility scenarios. Random Linear Network Coding (RLNC) adapts well to dynamic and distributed wireless topologies, while Physical Layer Network Coding (PLNC) extends this capability into the communication stack. Network coding enhances energy efficiency, security, and resource utilization, critical for next-generation wireless networks. It minimizes retransmissions and allows decentralized operation, aligning with the goals of ubiquitous, scalable, and resilient communication systems.

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